Normal Distribution

---

1. Normal Distribution of 1000 Random Values

A **normal distribution** is a continuous probability distribution that has a bell-shaped curve.

For a standard normal distribution:

- **Mean (μ) = 0**
- **Standard Deviation (σ) = 1**
- Number of observations = **1,000**

Since the data follows a standard normal distribution, the values are expected to be concentrated around the mean of **0**.

Expected Measures:

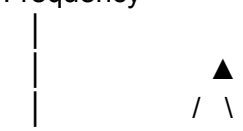
Measure	Approximate Value
Mean	0
Median	0
Mode	0

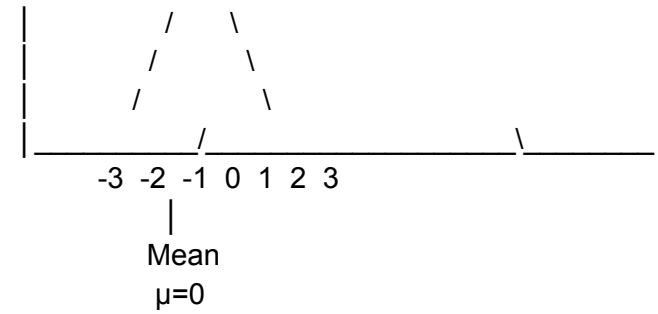
Note: Since the values are randomly generated, the actual mean and median may be slightly different from 0 in a sample of 1,000 observations.

---

2. Bell Curve of a Normal Distribution

A normal distribution has a **bell-shaped and symmetrical curve**.





Observation:

The highest point of the bell curve occurs around the **mean**.

For a standard normal distribution:

Mean = 0

The curve is symmetrical around the mean.

---

3. Percentage of Data Within 1, 2 and 3 Standard Deviations

The **Empirical Rule** states that for a normal distribution:

Within 1 Standard Deviation

Approximately **68%** of the data falls between:

$\mu-\sigma \quad \text{and} \quad \mu+\sigma$

For μ = 0 and σ = 1:

$-1 \leq X \leq 1$

Approximately 68% of data

Within 2 Standard Deviations

Approximately **95%** of the data falls between:

$\mu-2\sigma \quad \text{and} \quad \mu+2\sigma$

Therefore:

$-2 \leq X \leq 2$

Approximately 95% of data

Within 3 Standard Deviations

Approximately **99.7%** of the data falls between:

$\mu-3\sigma \quad \text{and} \quad \mu+3\sigma$

Therefore:

$-3 \leq X \leq 3$

Approximately 99.7% of data

Summary

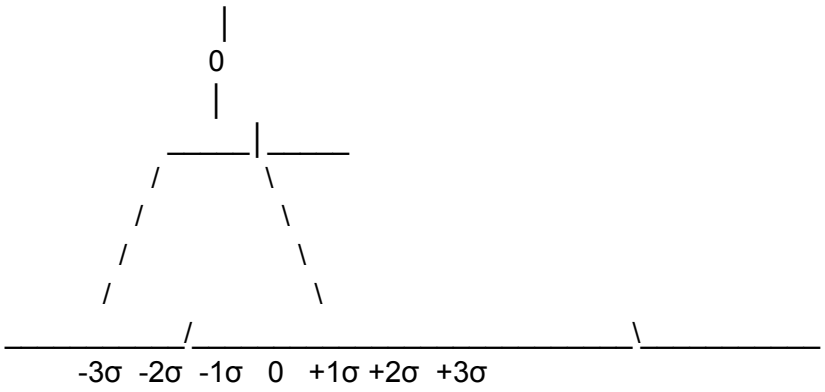
Range	Percentage of Data
Mean ± 1 Standard Deviation	**68%**
Mean ± 2 Standard Deviations	**95%**
Mean ± 3 Standard Deviations	**99.7%**

---

4. Shading the 68%, 95% and 99.7% Regions

On a normal bell curve, the regions can be represented as:

Mean



68% Region

The area between:

Mean − 1σ and Mean + 1σ

For the standard normal distribution:

−1 to +1

→ **Approximately 68%**

95% Region

The area between:

Mean − 2σ and Mean + 2σ

For the standard normal distribution:

−2 to +2

→ **Approximately 95%**

99.7% Region

The area between:

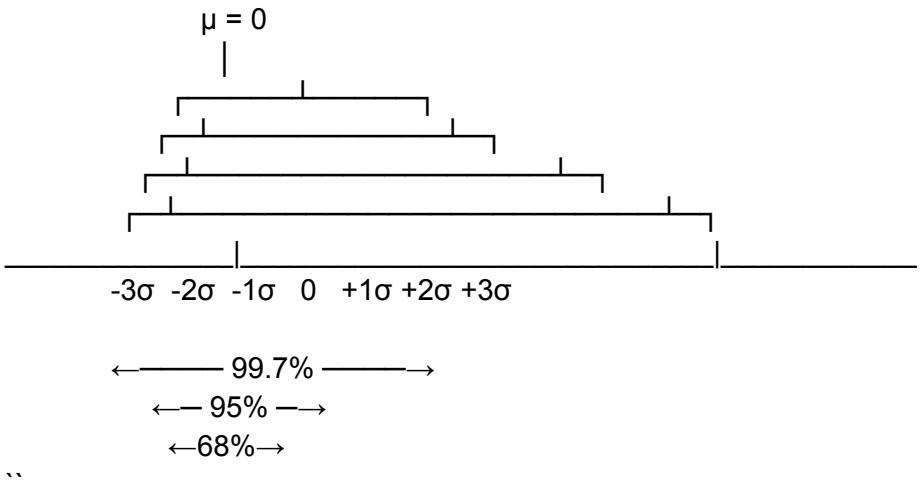
Mean − 3σ and Mean + 3σ

For the standard normal distribution:

−3 to +3

→ **Approximately 99.7%**

Visual Representation



``